

Ajinkya Ambadkar

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TECHNICAL SKILLS

Programming: Java, Python, C#, JavaScript, Typescript, Go, C, C++, SQL

Backend & Web Technologies: Spring Boot, REST APIs, API Gateway, Microservices Architecture, FastAPI, Event-Driven Architecture

Cloud Platforms & Messaging: AWS (Lambda, SNS, SQS, EventBridge, DynamoDB, EC2, S3), GCP (Pub/Sub, Compute Engine), Kafka

DevOps & Infrastructure: Docker, Kubernetes, Jenkins, Bamboo, Terraform, CI/CD Pipelines, PCF, Containerized Deployments

API & Integration: OpenAPI/Swagger, Contract-First Development, OAuth2, Enterprise Integration Patterns

Tools & Testing: JUnit, Mockito, Postman, Swagger/OpenAPI, SonarQube, Jira, Confluence, PyTorch, TDD, Git, Kibana, AppDynamics

EDUCATION

M.S. in Computer Science, Indiana University at Bloomington (GPA:3.93/4)

Aug 2024 – May 2026

EXPERIENCE

School of Public Health, Indiana University Bloomington | Research Assistant

Jun 2025 – Present

- Architected data engineering pipelines using Python and Scikit-learn for large-scale health data processing (1000+ records), implementing automated ETL workflows that reduced manual processing effort by 10+ hours/week.
- Designed and deployed interactive data visualization dashboard enabling stakeholders to perform real-time analysis of patient health trends and predictive markers.
- Established standardized data validation and delivery protocols, implementing quality assurance frameworks that reduced data errors by 20% for publication-grade research outputs.

Fidelity Information Services - FIS | Software Engineer II

Aug 2023 – Jun 2024

- Architected scalable API Gateway solutions using .NET and C#, implementing contract-first design principles to standardize API contracts across 30+ microservices, improving API consistency and reducing integration time by 40%.
- Designed and deployed event-driven microservices architecture leveraging asynchronous messaging patterns, boosting end-to-end performance by 30% and enabling parallel feature development across cross-functional teams.
- Optimized high-concurrency API workloads handling 500+ requests per second, reducing API response time by 25% through performance tuning, data contract optimization, and caching strategies.
- Modernized CI/CD pipelines using Jenkins and Git to enable incremental software delivery, achieving zero-downtime deployments and reducing production release cycle time by 35%.
- Spearheaded Agile delivery as Scrum Master for an 8-member cross-functional team, ensuring 100% on-time delivery of roadmap features through sprint planning, stakeholder collaboration, and continuous improvement.
- Engineered automated operational workflows across 30+ Windows services using .NET and WPF, streamlining system maintenance and reducing manual intervention by 60%.

Bitwise Solutions | Programmer Analyst

Oct 2020 – Aug 2023

- Architected and delivered 30+ enterprise-grade microservices using Spring Boot and Java for a major financial institution, designing robust REST APIs following OpenAPI specifications that scale in multi-tenant environments.
- Established API-first development practices across engineering teams, creating comprehensive API documentation and contract-first specifications that improved cross-team collaboration and reduced API integration defects by 35%.
- Implemented event-driven architecture using messaging patterns with GCP Pub/Sub and Kafka to decouple microservices, processing 100K+ messages daily while reducing application downtime by 20% through data-driven monitoring via Kibana and AppDynamics.
- Engineered high-performance microservices achieving 1,500 Transactions Per Second (TPS) using Apache JMeter for stress testing, ensuring SLA compliance for mission-critical financial APIs.
- Implemented comprehensive DevOps practices with 90%+ unit test coverage using JUnit and Mockito, integrating SonarQube into CI/CD pipelines for automated code quality and vulnerability detection.
- Managed complex data orchestration pipelines handling large datasets, resolving production data integrity issues through optimized SQL queries across MySQL and PostgreSQL databases.
- Served as Primary Technical Point of Contact (SPOC) during critical 3-month warranty support phase, providing 24/7 production oversight and troubleshooting for revenue-generating REST APIs (FeesAPI, DisputeRPT, AdjustedDisputeRPT).
- Strengthened API security by implementing AES-256 encryption and secure in-memory storage for sensitive data, ensuring zero data leakage during high-speed transmissions.
- Awarded Bitwise Excellence Award (August 2022) for delivering critical API enhancements and resolving production bottlenecks, boosting system availability by 25%.

PROJECTS

Distributed Chat System

- Architected a fault-tolerant distributed chat application using Flask, Redis Pub/Sub, and ZooKeeper on Kubernetes, achieving 99.9% message delivery through real-time leader election and service discovery.
- Integrated low-latency communication by orchestrating clustered message broadcasting across 4 pods, ensuring high availability and seamless failover during simulated peak traffic volumes.

IdeaGenie: Innovation Prioritization Platform

- Developed an end-to-end innovation-ranking platform using Flask backend and React frontend, exposing robust REST APIs following OpenAPI specifications to automate technical idea evaluation based on ROI and implementation constraints.
- Orchestrated a custom AI prioritization tool by fine-tuning a local LLM via Ollama, implementing Chain-of-Thought (CoT) reasoning to rank the top 3 innovation ideas with high transparency.